Introduction to Molecular Dynamics Simulations

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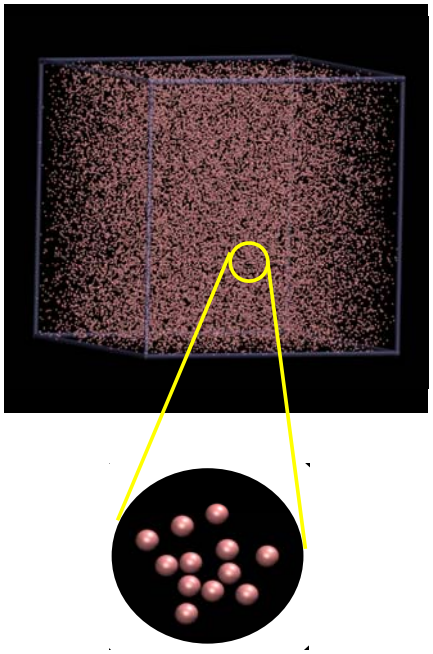
Newtonian equation of motion

- Total energy of the system

$$E = \sum_{i=1}^{N} \frac{1}{2} m_i v_i^2 + V(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N)$$

- Newtonian equation of motion

$$m_i \frac{d^2 \mathbf{r}_i}{dt^2} = -\nabla V(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N)$$
$$= F(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N)$$



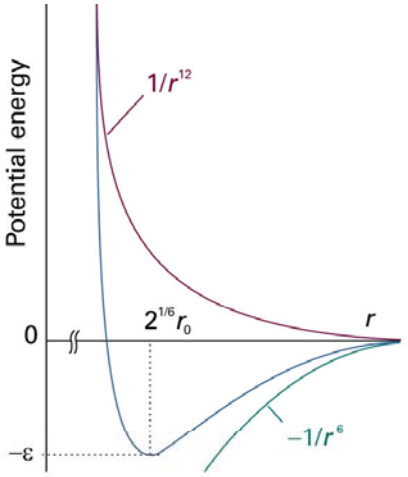
Potential energy function

- Lennard-Jones potential energy function for van der Waals interactions

$$V^{LJ}(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N) = \sum_{i=1}^{N} \sum_{j>i}^{N} 4\epsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^{6} \right]$$

  - Lorentz-Berthelot mixing rules for two different types atoms i, and j

$$\sigma_{ij} = \frac{\sigma_i + \sigma_j}{2}, \quad \epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j}$$



- Coulomb potential energy function for electrostatic interactions

$$V^{el}(\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_N) = \sum_{i=1}^{N} \sum_{j>i}^{N} \frac{1}{4\pi\epsilon_0} \frac{q_i q_j}{r_{ij}}$$

where $$\frac{1}{r_{ij}} = \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}$$

	ϵ (kcal/mol)	σ (Å)
He	0.02027	2.28
Ne	0.09340	2.72
Ar	0.2381	3.41
Kr	0.3259	3.83

Integrating Newtonian equations of motion -aelocity Verlet algorithm

- Taylor expansion

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)}{2!}h^2 + \frac{f^{(3)}(x)}{3!}h^3 + \cdots$$

- Advancing the position of a particle

$$\mathbf{r}_i(t+\delta t) = \mathbf{r}_i(t) + \delta t \mathbf{r}_i'(t) + \frac{1}{2}\delta t^2 \mathbf{r}_i''(t) + \cdots$$

$$\approx \mathbf{r}_i(t) + \delta t \mathbf{v}_i(t) + \frac{1}{2}\delta t^2 \mathbf{a}_i(t)$$

$$\approx \mathbf{r}_i(t) + \delta t \mathbf{v}_i(t) + \frac{1}{2}\delta t^2 \frac{\mathbf{F}_i(t)}{m_i}$$

$$\mathbf{r}_i(t+\delta t) = \mathbf{r}_i(t) + \delta t \mathbf{v}_i(t) + \frac{1}{2}\delta t^2 \frac{\mathbf{F}_i(t)}{m_i}$$

Integrating Newtonian equations of motion -velocity Verlet algorithm

- Advancing the velocity of a particle

$$\mathbf{v}_i(t+\frac{1}{2}\delta t) = \mathbf{v}_i(t) + \frac{1}{2}\delta t \mathbf{v}_i'(t)\delta t + \frac{1}{2}\delta t^2 \mathbf{v}_i''(t) + \cdots \approx \mathbf{v}_i(t) + \frac{1}{2}\delta t \mathbf{a}(t) \approx \mathbf{v}_i(t) + \frac{1}{2}\delta t \frac{\mathbf{F}_i(t)}{m_i}$$

$$\mathbf{v}_i(t+\frac{1}{2}\delta t) = \mathbf{v}_i(t+\delta t) - \frac{1}{2}\delta t \mathbf{v}_i'(t+\delta t) + \frac{1}{4}\delta t^2 \mathbf{v}_i''(t+\delta t) + \cdots$$

$$\approx \mathbf{v}_i(t+\delta t) - \frac{1}{2}\delta t \mathbf{a}(t+\delta t)$$

$$\approx \mathbf{v}_i(t+\delta t) - \frac{1}{2}\delta t \frac{\mathbf{F}_i(t+\delta t)}{m_i}$$

$$\mathbf{v}_i(t+\delta t) \approx \mathbf{v}_i(t+\frac{1}{2}\delta t) + \frac{1}{2}\delta t \frac{\mathbf{F}_i(t+\delta t)}{m_i}$$

$$\mathbf{v}_i(t+\delta t) = \mathbf{v}_i(t) + \frac{1}{2m_i}\delta t\left(\mathbf{F}_i(t) + \mathbf{F}_i(t+\delta t)\right)$$

Integrating Newtonian equations of motion -velocity Verlet algorithm

- Advancing the position of a particle

$$\mathbf{r}_i(t+\delta t) = \mathbf{r}_i(t) + \delta t \mathbf{v}_i(t) + \frac{1}{2}\delta t^2 \frac{\mathbf{F}_i(t)}{m_i}$$

- Advancing the velocity of a particle

$$\mathbf{v}_i(t+\delta t) = \mathbf{v}_i(t) + \frac{1}{2m_i}\delta t\left(\mathbf{F}_i(t) + \mathbf{F}_i(t+\delta t)\right)$$

Application of velocity Verlet algorithm to a harmonic oscillator model

- Newtonian equation of motion for a harmonic oscillator

$$m\frac{d^2x}{dt^2} = -kx \Rightarrow x_a(t) = A\cos(\omega t) + B\sin(\omega t) \qquad (\omega = \sqrt{k/m})$$

만약 $x(0) = 0,\ v(0) = v_0$ 라면,

$$x_a(t) = \frac{v_0}{\omega}\sin(\omega t)$$

- 예: $k = 4,\ m = 1,\ v_0 = 1$

$$x_a(t) = \frac{1}{2}\sin(2t), \qquad v_a(t) = \cos(2t)$$

$$x_i(t+\delta t) = x_i(t) + v_i(t)\delta t + \frac{1}{2}\frac{F_i(t)}{m_i}\delta t^2$$

$$v_i(t+\delta t) = v_i(t) + +\frac{1}{2m_i}\big(F_i(t) + F_i(t+\delta t)\big)\delta t$$

Maxwell-Botzmann distributions

- The Maxwell-Boltzmann distribution of molecular velocities in x direction

$$f(v_x)dv_x = \left(\frac{m}{2\pi k_B T}\right)^{1/2} \exp\left[-\frac{mv_x^2}{2k_B T}\right]dv_x$$

- The Maxwell-Boltzmann distribution of molecular velocities in three dimensions

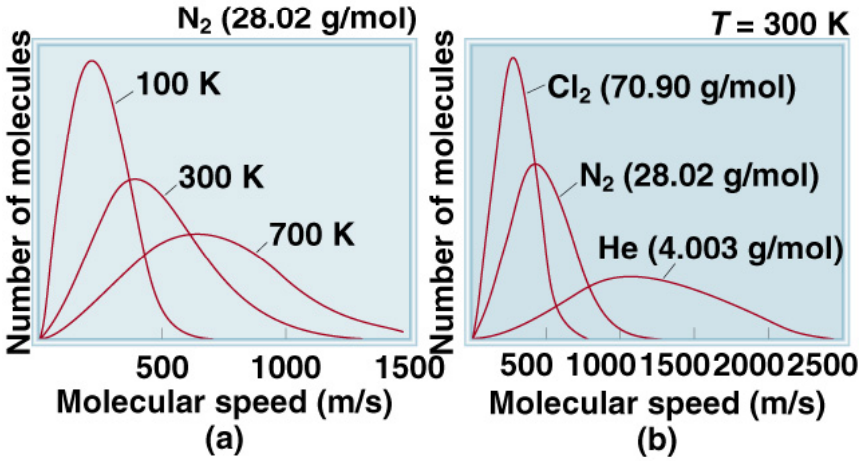
$$f(v_x)f(v_x)f(v_x)dv_x dv_y dv_z = \left(\frac{m}{2\pi k_B T}\right)^{3/2} \exp\left[-\frac{m(v_x^2+v_y^2+v_z^2)}{2k_B T}\right]dv_x dv_y dv_z$$

$$= \left(\frac{m}{2\pi k_B T}\right)^{3/2} \exp\left[-\frac{mv^2}{2k_B T}\right]dv_x dv_y dv_z$$

Maxwell distribution of speeds

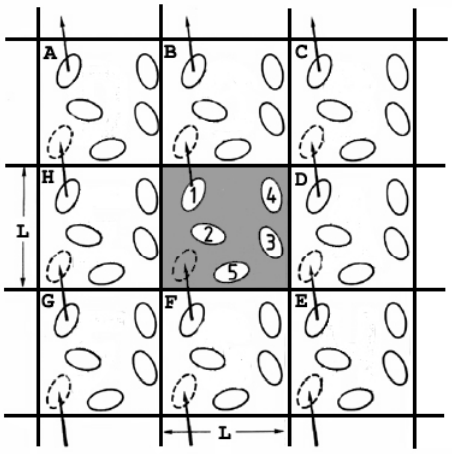
- The Maxwell distribution of speeds

$$f(v)dv = 4\pi\left(\frac{m}{2\pi k_B T}\right)^{3/2} v^2 \exp\left[-\frac{mv^2}{2k_B T}\right]dv$$



Periodic boundary conditions and minimum image convention

- Periodic boundary conditions (PBC)
- Minimum Image Convention (MIC)



Kinetic model of gases

- The total momentum change during Δt

$$\text{momentum change} = \text{impact} = \frac{NAv_x\Delta t}{2V} \times 2mv_x$$

$$= \frac{NAmv_x^2\Delta t}{V}$$

- Average pressure

$$P = \frac{\text{impact}}{A\Delta t} = \frac{Nm\langle v_x^2\rangle}{V}$$

- Average kinetic energy of a molecule in *x* direction

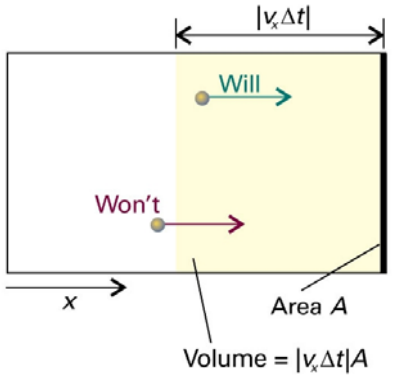


Fig. 21.2 A molecule will reach the wall on the right within an interval Δt if it is within a distance $v_x\Delta t$ of the wall and travelling to the right.

$$\text{average KE of a molecule} = \frac{1}{2}m\langle v_x^2\rangle = \frac{1}{2}\frac{PV}{N} = \frac{1}{2}\frac{Nk_BT}{N} = \frac{1}{2}k_BT$$

Velocity scaling

- Average kinetic energy of a molecule

$$\text{average KE of a molecule} = \frac{1}{2}m\langle v^2\rangle = \frac{1}{2}m\langle v_x^2 + v_y^2 + v_z^2\rangle$$

$$= \frac{1}{2}m\langle v_x^2\rangle + \frac{1}{2}m\langle v_y^2\rangle + \frac{1}{2}m\langle v_z^2\rangle = \frac{1}{2}k_BT + \frac{1}{2}k_BT + \frac{1}{2}k_BT = \frac{3}{2}k_BT$$

- Scaling factor

$$\frac{1}{2}m\sum_{n=1}^{N}\langle (v_n^i)^2\rangle = \frac{3}{2}(N-1)k_BT$$

$$\frac{1}{2}m\sum_{i=1}^{N}\langle (v_n^f)^2\rangle^f = \frac{3}{2}(N-1)k_BT_{ref}$$

$$\text{scaling factor} = \left(\frac{3(N-1)T_{ref}}{m\sum_{n=1}^{N}\langle (v_n^i)^2\rangle}\right)^{1/2}$$

$$\sum_{n=1}^{N}\langle (v_n^f)^2\rangle = \frac{(N-1)T_{ref}}{(N-1)T}\sum_{n=1}^{N}\langle (v_n^i)^2\rangle$$

$$= \sum_{n=1}^{N}\left\langle \left\{\left(\frac{(N-1)T_{ref}}{(N-1)T}\right)^{1/2} v_n^i\right\}^2\right\rangle$$

$$= \sum_{i=1}^{N}\left\langle \left\{\left(\frac{3(N-1)T_{ref}}{m\sum_{n=1}^{N}\langle (v_n^i)^2\rangle}\right)^{1/2} v_n^i\right\}^2\right\rangle$$

Calculation of pressure

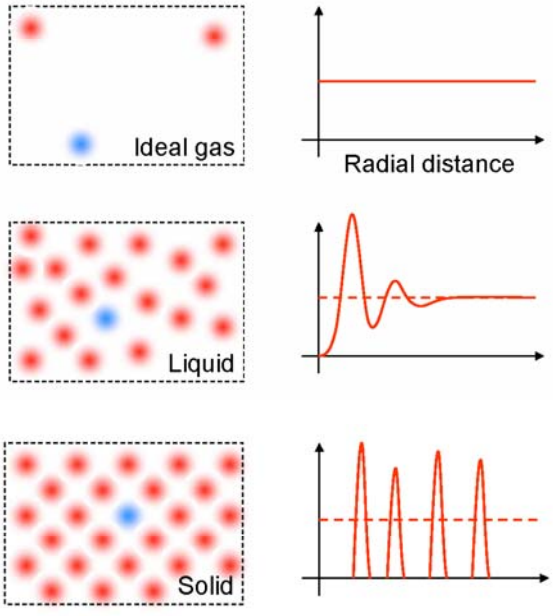
- Virial theorem for pressure

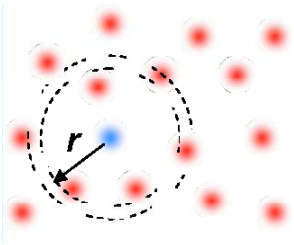
$$P = \rho k_B T + \frac{1}{3V}\left\langle \sum_{i=1}^{N}\sum_{j>i}^{N} \mathbf{r}_{ij} \cdot \mathbf{F}_{ij} \right\rangle$$

where $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ and $\mathbf{F}_{ij}$ is the force acting on particle *i* by particle *j*.

Radial distribution function - structured fluids

- Radial distribution function of gas, liquid, and solid





Calculate the number of particles inside a radial segement ***(r,r+Δr)*** from one specific particle in the system.

Average the distribution over all particles in the system.

Divide the (average) number of particles in a segment by the segment volume.

Finally, normalize thus obtained radial density profile with the bulk density ***N/V*** of the system.